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Workshop

to use an additional framework that supports grids (like Bootstrap) if you want that.

MDL also uses a CSS class naming methodology called BEM that makes it easy to understand how to use a class just by the way it's named. Let's see how this works with something as simple as a button before we go into what BEM is.

Let's say we want to recreate that 'SEND' button from the email compose windows in Inbox. We'll first need to start with a plain old button:

```
<button>Send</button> 
```

To which we add the 'mdl-button' class:

```
<button class="mdl-button">Send</button> 
```

Now this looks more like something we'd see as a button in an MD app, but our button needs to be raised so we add a class for that:

```
<button class="mdl-button mdl-button--raised">Send</button> 
```

Better but our needs to be coloured blue:

```
<button class="mdl-button mdl-button--raised mdl-button--colored">Send</button> 
```

Note that this colouring will only appear if it is the primary theme colour. Otherwise the button will be of the primary theme colour. Now if we also wanted this to show that ripple animation when tapped, we would just need to add the appropriate classes ('mdl-js-button' and 'mdl-js-ripple-effect').

You must have noticed some kind of pattern with these CSS classes, especially with the use of double hyphens. That is because, as previously mentioned, MDL uses a naming methodology called BEM, which we'll look into next.

## BEM in MDL Through Examples

As layouts become increasingly complex with layers of HTML tags each with different CSS classes coming together, to give a single effect or make a single component, it can be confusing to know how those classes relate to each other. Which is the class for the main parent component? Which is for the sub component? Wait,

where do you add what class to make the header blue?

BEM which stands for Block, Element, Modifier, is a style of naming CSS classes that tried to reduce that confusion by making the purpose and use of a class clear from its name.

In the previous example, 'mdl-button' represents a block, i.e. a high-level component within our code, in this case a button. The 'mdl' preceding the button ensures that MDL buttons don't clash with some other framework that also happens to use a class called 'button'.

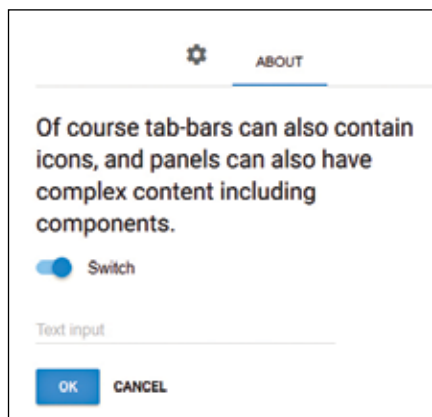
In 'mdl-button--raised' we have a modifier applied to 'mdl-button'. Anywhere you see something like 'xyz-block-modifier' you will know that the class is modifying the base block / component in some way.

The 'Elements' bit of BML is something we didn't see in the button example since it is a rather simple one. However, imagine a more complex component with many subparts, where each of them requires us to set a class in order for the component to work as a whole.

A good example of this is the MDL tabs layout. While the major component / layout is a tab panel, it is composed of elements such as tab buttons and content panels that are linked to the different tabs. There is where the Element syntax of BEM comes in.

The structure of the tab component is as follows:

```
<div class="mdl-tabs mdl-js-tabs">
  <div class="mdl-tabs__tab-
bar">
```



We have expanded the contents within these tabs to make them more clear for print.

```
<a href="#tab1-panel"
class="mdl-tabs__tab is-active">Tab
1</a>

<a href="#tab2-panel"
class="mdl-tabs__tab">Tab 2</a>
</div>

<div class="mdl-tabs__panel
is-active" id="tab1-panel">
  Tab 1 contents
</div>
<div class="mdl-tabs__panel"
id="tab2-panel">
  Tab 2 contents
</div>
</div>
```

Here, the main component is 'mdl-tabs' (the 'mdl-js-tabs' is required in addition to make it function). The 'div' with the 'mdl-tabs\_\_tab-bar' class is an element of the tabbed panel component. As you can see the CSS class for this tab bar is the CSS class name of the main component followed by two underscores and then the name of the element of that component. Further within the 'tab-bar' element are the 'tab' elements that represent the actual tabs on top of the screen. The tabs component also has two additional child elements that are both panel components of tabs.

With a clear naming pattern like this you know in advance where a particular CSS class is applicable. For instance, while a 'panel' class might apply in many cases and could cause confusion, here it is quite clear the 'mdl-tabs\_\_panel' is only a CSS class that would be used inside an 'mdl-tabs' component.

MDL likewise has other layouts and components that are composed of elements.

## Conclusion

MDL is the latest entry in list of web design frameworks; there seems to be a new one every other week. However, Material Design is quite an important consideration now if you want your app or website to fit in on Android devices.

Google has also made available a couple of templates that use MDL in a blog, dashboard or article. Also included is a skin for Android.com using MDL. 